MIL-F-14187D
30 June 1980
SUPERSEDING
MIL-F-14187C
16 September 1965

MILITARY SPECIFICATION

FASTENER, REFRIGERATOR PANEL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type and size of fastener for connecting wall, floor, and roof panels of a prefabricated refrigerator.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-P-1183	-	Plastic Molding Material, Acrylonitrile - Butadiene - Styrene, (ABS) Rigid.
QQ-A-250/11	-	Aluminum Alloy 6061, Plate and Sheet.
QQ-A-591	-	Aluminum Alloy Die Castings.
QQ-Z-325	-	Zinc Coating, Electrodeposited, Requirements For.
VV-G-671	-	Grease, Graphite.
PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-B-676	-	Boxes, Set-Up, Paperboard.

MILITARY

MIL-R-3065	-	Rubber, Fabricated Products.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5325

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:55Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-147 - Palletized and Containerized Unit Loads 40" x 48" Pallets, Skids, Runners or Pallet-Type Base.
MIL-STD-417 - Rubber Compositions, Vulcanized, General Purpose, Solid (Symbols and Tests).
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment.

DRAWINGS

US ARMY NATICK RESEARCH AND DEVELOPMENT COMMAND

5-13-1858 - Fastener, Refrigerator Panel; Assembly and Installation.
5-13-1859 - Fastener, Refrigerator Panel; Details.
5-13-1860 - Fastener, Refrigerator Panel; Details.

(Miniature copy of Drawing 5-13-1858, identified as figure 1, is attached for information purposes only.)

(Copies of specifications, standards, and drawings, required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 86 - Zinc Alloy Die Castings

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

Source: https://assist.dla.mil -- Downloaded: 202610-18T02:55Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Materials. The materials specified shall conform to the applicable specifications and requirements specified hereinafter. Where materials are not definitely specified, they shall be of the quality normally used by the contractor in fabricating his fasteners. Unless otherwise specified herein, commercial tolerances shall apply to all metal thicknesses and diameters.

3.1.1 Aluminum.

3.1.1.1 Aluminum alloy 6061 plate and sheet. Aluminum alloy 6061 plate and sheet shall conform to QQ-A-250/11.

3.1.1.2 Aluminum alloy die castings. Aluminum alloy die castings shall conform to alloy No. 380 of QQ-A-591.

3.1.2 Zinc-base alloy die castings. Zinc-base alloy die castings shall conform to ASTM B 86.

3.1.3 Rubber. Molded or extruded rubber shall conform to type, class, grade, and suffix RS610AF1 of MIL-STD-417 and MIL-R-3065.

3.1.4 Plastic. The plastic material shall conform to type II of L-P-1183.

3.1.5 Graphite grease. Graphite grease shall conform to VV-G-671.

3.2 Design and construction. The design and construction of the refrigerator panel fastener and component parts covered by this specification shall conform to the requirements herein and the design, details, dimensions, and material classifications shown or specified on the drawings listed in 2.1. The panel fastener shall consist of a hook assembly and an eye assembly. The hook assembly shall contain a hook, which when rotated by a lever turning clockwise shall engage the eye. The lever shall be turned manually by 5/16-inch hexagonal wrench. The lever shall also keep the hook positively engaged with an eye. A reverse rotation shall disengage the hook from the eye and return it to a retracted position where it shall be retained by a keeper.

3.2.1 Hook assembly.

3.2.1.1 Hook. The hook shall be fabricated of the aluminum alloy sheet specified in 3.1.1.1.

3.2.1.2 Lever and lever gate. The lever and lever gate shall be fabricated of the zinc-base alloy specified in 3.1.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:55Z
Check the source to verify that this is the current version before use.

3.2.1.3 Guide pins. Guide pins shall be fabricated of yellow brass.

3.2.1.4 Keeper. The keeper shall be fabricated of the rubber specified in 3.1.3.

3.2.1.5 Hook assembly housing. The hook assembly housing shall be fabricated of the aluminum alloy or zinc-base alloy specified in 3.1.1.2 or 3.1.2.

3.2.2 Eye assembly.

3.2.2.1 Eye. The eye shall be fabricated from spring steel rod as shown on Drawing 5-13-1860. The eye shall be constructed so that when a force of 300 to 350 pounds is applied to the eye, the loop of the eye shall not be displaced more than 1/8 inch when it is in its normal assembled position when tested as specified in 4.3.1. When displaced a maximum of 1/4 inch, the loop of the eye will take not more than a 1/16 inch permanent set when tested as specified in 4.3.1. The eye shall withstand an applied flexing force of not less than 700 pounds without fracture when tested as specified in 4.3.1.

3.2.2.2 Eye cover. The eye cover shall be fabricated from the plastic material specified on Drawing 5-13-1860.

3.2.3 Assembly. The refrigerator panel fastener shall be assembled as shown on Drawing 5-13-1858. Graphite grease as specified in 3.1.5 shall be applied to moving parts at time of assembly. The assembled fastener shall operate without breakdown when the hook is moved from its retracted position to its engaged position and back 50 times when tested as specified in 4.3.2.

3.3 Finish. All metal components of the refrigerator panel fasteners, except the eye, shall have the natural metal finish. The eye shall be zinc plated conforming to type II, class 3 of QQ-Z-325.

3.4 Marking for identification. Panel fasteners shall be marked for identification in accordance with MIL-STD-130.

3.5 Workmanship. All sharp edges, burrs or slivers shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:55Z
Check the source to verify that this is the current version before use.

4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.2 End item inspection. The inspection lot shall consist of all panel fasteners offered for inspection at one time. The sample unit shall be one completely fabricated panel fastener.

4.2.2.1 Visual examination. The completely assembled panel fastener shall be examined for the defects in table I. The inspection level shall be II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Metal components, except the eye, not natural finish.		X
	Eye not zinc plated.	X	
	Sharp edges, slivers or burrs.	X	
Design, construction and workmanship general (applicable to all components and assemblies)	Any component missing, fractured, malformed, dented, sprung or warped.	X	
Marking for identification	Missing, incomplete, not legible.		X

4.2.2.2 Dimensional examination. Examination shall be made of the refrigerator panel fastener to determine compliance with dimensional requirements specified on the drawings. Any deviation from the specified requirements shall constitute a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

Source: https://assist.dla.mil -- Downloaded: 202610-18T02:55Z
Check the source to verify that this is the current version before use.

4.2.2.3 Testing of the end item. The end item shall be tested as specified in 4.3.1 and 4.3.2. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.2.3 Packaging inspection. An examination shall be made to determine whether the preservation, packing, and marking comply with the section 5 requirement. Defects shall be scored as set forth in table II. The lot shall be the number of containers offered for inspection at one time. The sample unit shall be one container fully packaged except it need not be closed. The inspection level shall be S-2 and the AQL shall be 4.0 defects, expressed in terms of defects per hundred units.

TABLE II. Packaging defects

Examine	Defect
Marking, exterior and interior	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.
Material	Component missing, damaged, or otherwise defective.
Content	Number per container is more or less than required.

4.2.3.1 Examination of palletization. An examination shall be made to determine that palletization complies with section 5 requirements. Defects shall be as listed in table III. The sample unit for this examination shall be one palletized unit load ready for shipment. The lot shall be the number of palletized unit loads offered for inspection at one time. The inspection level shall be S-1 with an AQL of 6.5 expressed in terms of defects per hundred units.

TABLE III. Examination of palletization

Examine	Defect
Finished dimension	Length, width or height exceeds specified maximum requirement.
Palletization	Not as specified. Pallet pattern not as specified. Load not bonded with required strap specified.
Weight	Exceeds maximum load limits.
Marking	Omitted, incorrect, illegible or improper size, location, sequence or method of application.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:55Z
Check the source to verify that this is the current version before use.

4.3 Methods of testing.

4.3.1 Eye strength. A force of 300 to 350 pounds shall be applied to the loop of the sample refrigerator panel fastener eye and the displacement of the eye measured to determine compliance with 3.2.2.1. The force shall be increased until the loop of the eye is displaced 1/4 inch. The force shall then be removed and the amount of permanent set measured to determine compliance with 3.2.2.1. A flexing force of 700 pounds shall be applied to determine compliance with the fracture requirements of 3.2.2.1.

4.3.2 Operation test. The sample refrigerator panel fastener shall be assembled in a mock-up to simulate actual use. The fastener shall then be operated by a 5/16 inch hexagonal wrench so the hook is moved from the retracted position to the engaged position and back 50 times. The fastener shall be examined for compliance with 3.2.3 during and after the test.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Each complete refrigerator panel fastener shall be unit packed in the most compact manner in a snug-fitting set-up paperboard or folding paperboard box conforming to type I, variety plain, class A, style 3 or 4 of PPP-B-676, or variety 1, style III or VIII, type G, class 1 of PPP-B-566, respectively. Each box shall be closed in accordance with the appendix of the applicable specification.

5.1.2 Commercial. Refrigerator panel fasteners shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Fifty refrigerator panel fasteners, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, V2s of PPP-B-636. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V and reinforced as specified in the appendix of PPP-B-636.

5.2.2 Level B packing. Fifty refrigerator panel fasteners, preserved as specified in 5.1, shall be packed in a snugfitting fiberboard shipping container conforming to style RSC, type CF, variety option, or type SF, class domestic, grade 275 of PPP-B-636. Closure shall be in accordance with method II as specified in the appendix of PPP-B-636.

Source: https://assist.dla.mil -- Downloaded: 202610-18T02:55Z
Check the source to verify that this is the current version before use.

5.2.2.1 Weather-resistant containers. When specified (see 6.2), the shipping container shall be of a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636.

5.2.3 Commercial packing. Refrigerator panel fasteners, preserves as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Palletization. When specified (see 6.2), refrigerator panel fasteners packed as specified shall be palletized in accordance with Load Type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147; the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking. In addition to any special marking required by the contract, unit packs and shipping containers shall be marked in accordance with the requirements of MIL-STD-129 or MIL-STD-1188, as applicable.

6. NOTES

6.1 Intended use. The refrigerator panel fastener is intended to be used as a component of the wall, floor and roof panels of pre-fabricated refrigerators meeting the requirements of MIL-R-10932, to join the panels when the refrigerators are erected.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When palletization is required (see 5.3).
(c) Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
(d) Whether weather-resistant class fiberboard shipping containers are required for level B shipment (see 5.2.2.1).

6.3 Changes from the previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:

Army - GL
Air Force - 99

Preparing activity:

Army - GL

Project No. 5325-0127

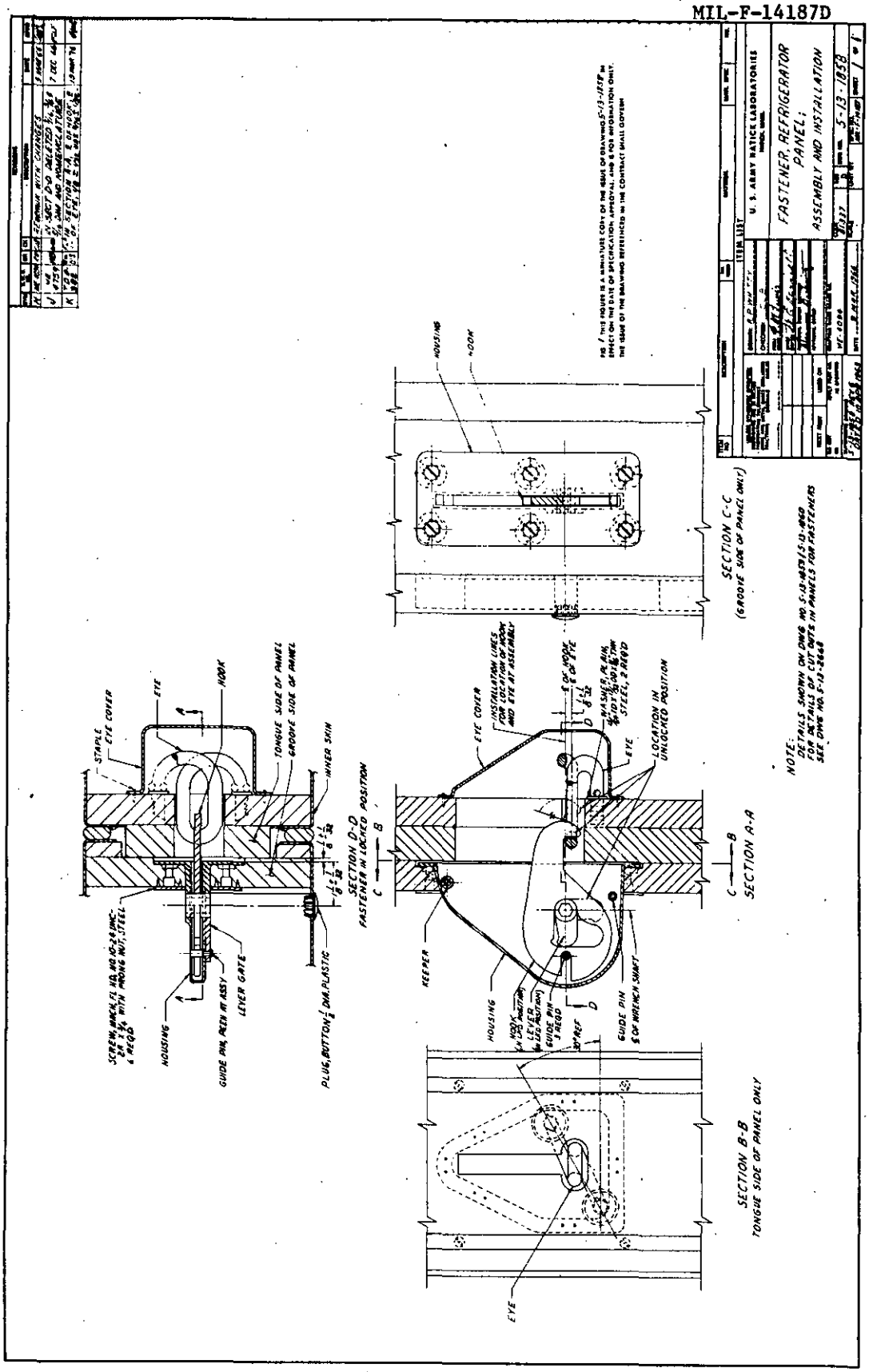
Review activity:

DLA - IS

User activities:

Army - MD
Navy - MC

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:55Z
Check the source to verify that this is the current version before use.



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:55Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-F-14187D; Fastener, Refrigerator Panel

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426

EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:55Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
US ARMY NATICK RESEARCH and DEVELOPMENT COMMAND
NATICK, MASSACHUSETTS 01760
OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300
DRDNA-ES

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD-314



Commander
US Army Natick Research and Development Command
ATTN: DRDNA-ES
Natick, MA 01760

NATICK OP 1 AUG 78
FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:55Z
Check the source to verify that this is the current version before use.